

# Pinned group summary: $\text{SL}(\mathbf{n}=4)$

Pinning-Sympy automated output

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1 Basic group attributes

|                    |                                    |
|--------------------|------------------------------------|
| Group              | SL (n=4)                           |
| Matrix size        | $4 \times 4$                       |
| Rank               | 3                                  |
| Defining equations | <i>DefiningEquationPlaceholder</i> |

1.1 Generic Lie algebra element

$$\begin{bmatrix} x_{00} & x_{01} & x_{02} & x_{03} \\ x_{10} & x_{11} & x_{12} & x_{13} \\ x_{20} & x_{21} & x_{22} & x_{23} \\ x_{30} & x_{31} & x_{32} & -x_{00} - x_{11} - x_{22} \end{bmatrix}$$

1.2 Generic torus element

$$\begin{bmatrix} t_0 & 0 & 0 & 0 \\ 0 & t_1 & 0 & 0 \\ 0 & 0 & t_2 & 0 \\ 0 & 0 & 0 & \frac{1}{t_0 t_1 t_2} \end{bmatrix}$$

1.3 Trivial characters

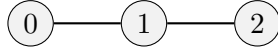
$$\begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix}$$

## 2 Root system

### Summary Properties

|                 |      |
|-----------------|------|
| Dynkin type     | A3   |
| Irreducible     | True |
| Reduced         | True |
| Simply laced    | True |
| Number of roots | 12   |

### 2.1 Dynkin diagram



### 2.2 Cartan matrix

$$\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

### 2.3 Roots and coroots

| Root            | Coroot          | Simple | Sign | Multipliable |
|-----------------|-----------------|--------|------|--------------|
| $(-1, 0, 0, 1)$ | $(-1, 0, 0, 1)$ | No     | —    | No           |
| $(-1, 0, 1, 0)$ | $(-1, 0, 1, 0)$ | No     | —    | No           |
| $(-1, 1, 0, 0)$ | $(-1, 1, 0, 0)$ | No     | —    | No           |
| $(0, -1, 0, 1)$ | $(0, -1, 0, 1)$ | No     | —    | No           |
| $(0, -1, 1, 0)$ | $(0, -1, 1, 0)$ | No     | —    | No           |
| $(0, 0, -1, 1)$ | $(0, 0, -1, 1)$ | No     | —    | No           |
| $(0, 0, 1, -1)$ | $(0, 0, 1, -1)$ | Yes    | +    | No           |
| $(0, 1, -1, 0)$ | $(0, 1, -1, 0)$ | Yes    | +    | No           |
| $(0, 1, 0, -1)$ | $(0, 1, 0, -1)$ | No     | +    | No           |
| $(1, -1, 0, 0)$ | $(1, -1, 0, 0)$ | Yes    | +    | No           |
| $(1, 0, -1, 0)$ | $(1, 0, -1, 0)$ | No     | +    | No           |
| $(1, 0, 0, -1)$ | $(1, 0, 0, -1)$ | No     | +    | No           |

### 2.4 Linear combinations of roots

**Definition 2.1.** For two roots  $\alpha, \beta$  in a root system  $\Phi$ , the set of positive integral linear combinations of  $\alpha, \beta$  that are also roots is denoted  $(\alpha, \beta) = \{i\alpha + j\beta \in \Phi : i, j \in \mathbb{Z}_{\geq 1}\}$

| $\alpha$        | $\beta$         | Pairs $(i, j)$ |
|-----------------|-----------------|----------------|
| $(-1, 0, 0, 1)$ | $(0, 0, 1, -1)$ | $(1, 1)$       |
| $(-1, 0, 0, 1)$ | $(0, 1, 0, -1)$ | $(1, 1)$       |
| $(-1, 0, 0, 1)$ | $(1, -1, 0, 0)$ | $(1, 1)$       |
| $(-1, 0, 0, 1)$ | $(1, 0, -1, 0)$ | $(1, 1)$       |
| $(-1, 0, 1, 0)$ | $(0, 0, -1, 1)$ | $(1, 1)$       |
| $(-1, 0, 1, 0)$ | $(0, 1, -1, 0)$ | $(1, 1)$       |
| $(-1, 0, 1, 0)$ | $(1, -1, 0, 0)$ | $(1, 1)$       |
| $(-1, 0, 1, 0)$ | $(1, 0, 0, -1)$ | $(1, 1)$       |
| $(-1, 1, 0, 0)$ | $(0, -1, 0, 1)$ | $(1, 1)$       |
| $(-1, 1, 0, 0)$ | $(0, -1, 1, 0)$ | $(1, 1)$       |
| $(-1, 1, 0, 0)$ | $(1, 0, -1, 0)$ | $(1, 1)$       |
| $(-1, 1, 0, 0)$ | $(1, 0, 0, -1)$ | $(1, 1)$       |
| $(0, -1, 0, 1)$ | $(0, 0, 1, -1)$ | $(1, 1)$       |
| $(0, -1, 0, 1)$ | $(0, 1, -1, 0)$ | $(1, 1)$       |
| $(0, -1, 0, 1)$ | $(1, 0, 0, -1)$ | $(1, 1)$       |
| $(0, -1, 1, 0)$ | $(0, 0, -1, 1)$ | $(1, 1)$       |
| $(0, -1, 1, 0)$ | $(0, 1, 0, -1)$ | $(1, 1)$       |
| $(0, -1, 1, 0)$ | $(1, 0, -1, 0)$ | $(1, 1)$       |
| $(0, 0, -1, 1)$ | $(0, 1, 0, -1)$ | $(1, 1)$       |
| $(0, 0, -1, 1)$ | $(1, 0, 0, -1)$ | $(1, 1)$       |
| $(0, 0, 1, -1)$ | $(0, 1, -1, 0)$ | $(1, 1)$       |
| $(0, 0, 1, -1)$ | $(1, 0, -1, 0)$ | $(1, 1)$       |
| $(0, 1, -1, 0)$ | $(1, -1, 0, 0)$ | $(1, 1)$       |
| $(0, 1, 0, -1)$ | $(1, -1, 0, 0)$ | $(1, 1)$       |

### 3 Root spaces

| Root            | Dimension |
|-----------------|-----------|
| $(-1, 0, 0, 1)$ | 1         |
| $(-1, 0, 1, 0)$ | 1         |
| $(-1, 1, 0, 0)$ | 1         |
| $(0, -1, 0, 1)$ | 1         |
| $(0, -1, 1, 0)$ | 1         |
| $(0, 0, -1, 1)$ | 1         |
| $(0, 0, 1, -1)$ | 1         |
| $(0, 1, -1, 0)$ | 1         |
| $(0, 1, 0, -1)$ | 1         |
| $(1, -1, 0, 0)$ | 1         |
| $(1, 0, -1, 0)$ | 1         |
| $(1, 0, 0, -1)$ | 1         |

RootSpaceTablePlaceholder

### 4 Root subgroups

RootSubgroupTablePlaceholder

HomomorphismDefectTablePlaceholder

HomomorphismDefectIdentitiesPlaceholder

## 5 Commutators

CommutatorCoefficientTablePlaceholder  
CommutatorIdentitiesPlaceholder

## 6 Weyl group

WeylGroupPlaceholder  
WeylConjugationCoefficientPlaceholder  
WeylConjugationIdentitiesPlaceholder

## 7 Coroot torus elements ( $h_\alpha$ )

CorootTorusElementPlaceholder